

CERTIFIED NONEXTENSION OF A HADAMARD PAIR TO A MUB TRIPLE IN DIMENSION SIX

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ABSTRACT. We give a validated computer-assisted proof that an explicit member of the Matolcsi–Szöllősi non-affine Hadamard family of order six cannot be completed with the standard basis to a triple of mutually unbiased bases. After fixing global phase, common unbiased rays are the zeros of five real analytic equations on a five-dimensional torus. A complete interval solve isolates exactly 52 rays. Outward-rounded interval evaluation of all pairwise inner products produces a conservative possible-orthogonality graph with 48 edges and no triangle. Since the six columns of a third basis would form a six-clique, no such basis exists. As a complementary breaker result, we exhibit an exact third basis at the endpoint parameter $x = -1$, showing why randomized parameter searches cannot by themselves decide isolated cases.

1. INTRODUCTION

Two orthonormal bases $\mathcal{B}$ and $\mathcal{C}$ of $\mathbb{C}^d$ are mutually unbiased if

$$|\langle b, c \rangle|^2 = \frac{1}{d} \quad (b \in \mathcal{B}, c \in \mathcal{C}).$$

The maximum number of mutually unbiased bases (MUBs) is $d+1$ in prime-power dimensions, whereas even the existence of four MUBs in dimension six remains open. The recent review of McNulty and Weigert records this as a central open problem and explicitly asks for rigorous nonextension results for selected order-six Hadamard pairs [6].

A dephased complex Hadamard matrix H of order six gives the pair $\{I_6, H/\sqrt{6}\}$. Matolcsi and Szöllősi constructed a symmetric non-affine one-parameter family $M_6(x)$ [5]. Numerical work subsequently explored MUB extensions of this and other non-affine families, but expressly used a randomized nondecision procedure [2]. The purpose of this paper is to replace numerical absence at one exact parameter by a complete certificate.

Our principal result is the following.

Theorem 1.1. *Let K be the explicit matrix in (1). Then the mutually unbiased pair $\{I_6, K/\sqrt{6}\}$ does not extend to a triple of mutually unbiased*

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bases in $\mathbb{C}^6$. In fact, among all rays unbiased to both bases, no three are pairwise orthogonal.

The proof has two finite parts. First, a complete validated interval solve isolates all 52 common-unbiased rays. Second, interval evaluation of every pairwise inner product constructs a supergraph of the exact orthogonality graph. This supergraph is triangle-free. The supergraph convention is deliberately one-sided: an edge is retained whenever interval arithmetic cannot exclude orthogonality. Consequently no approximate zero is ever used as a proof of nonorthogonality.

The theorem concerns one explicit pair, not the entire Matolcsi–Szöllősi family and not the general dimension-six MUB problem. The computation is a validated interval proof with a small independent postprocessor; no claim of proof-assistant formalization is made.

2. THE EXACT HADAMARD MATRIX

Put

$$\xi = e^{\pi i/3} = \frac{1 + i\sqrt{3}}{2}$$

and define the six unimodular algebraic numbers

$$\begin{aligned} a &= \frac{(-5 - i\sqrt{3})(1 - 3i/\sqrt{7})}{8}, & d &= \frac{(-5 - i\sqrt{3})(1 + 3i/\sqrt{7})}{8}, \\ b &= -\frac{\xi(1 + i\sqrt{15})}{4}, & c &= -\frac{\xi(1 - i\sqrt{15})}{4}, \\ p &= \frac{(-1 + 3i\sqrt{3})(1 + 3i/\sqrt{7})}{8}, & q &= \frac{(-1 + 3i\sqrt{3})(1 - 3i/\sqrt{7})}{8}. \end{aligned}$$

The target matrix is

$$(1) \quad K = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & -1 & \xi & \xi & -\xi & -\xi \\ 1 & \xi & d & a & b & c \\ 1 & \xi & a & d & c & b \\ 1 & -\xi & b & c & p & q \\ 1 & -\xi & c & b & q & p \end{pmatrix}.$$

These are the formulas of the family $M_6(x)$ at $x = \xi$, with a fixed choice of the paired roots [5, Sec. 3]. Direct arithmetic in $\mathbb{Q}(i\sqrt{3}, i\sqrt{7}, i\sqrt{15})$ gives

$$(2) \quad |K_{jk}| = 1, \quad KK^* = 6I_6.$$

Thus $K/\sqrt{6}$ is unitary and its columns form a basis mutually unbiased to the standard basis.

3. COMMON-UNBIASED RAYS

Let v be a unit vector unbiased to the standard basis. None of its coordinates vanishes, and after fixing global phase it has the unique form

$$(3) \quad v(\theta) = \frac{1}{\sqrt{6}}(1, e^{i\theta_1}, \dots, e^{i\theta_5})^T, \quad \theta \in [0, 2\pi)^5.$$

It is also unbiased to $K/\sqrt{6}$ precisely when

$$(4) \quad \left| \sum_{j=0}^5 \overline{K_{jk}} e^{i\theta_j} \right|^2 = 6, \quad 0 \leq k \leq 5, \text{quad} \theta_0 = 0.$$

Lemma 3.1. *In (4), any five of the six intensity equations imply the remaining equation.*

Proof. With $z = (1, e^{i\theta_1}, \dots, e^{i\theta_5})^T$, identity (2) gives

$$\sum_{k=0}^5 |(K^* z)_k|^2 = \|K^* z\|_2^2 = 6\|z\|_2^2 = 36.$$

If five summands equal 6, the sixth also equals 6. $\square$

For $0 \leq k \leq 4$, define

$$(5) \quad F_k(\theta) = \left| \sum_{j=0}^5 \overline{K_{jk}} e^{i\theta_j} \right|^2 - 6, \quad F = (F_0, \dots, F_4).$$

Lemma 3.1 reduces the classification to the square system $F(\theta) = 0$ on the full phase cube.

4. VALIDATED ENUMERATION

We used the complete and validated IbexSolve implementation in IBEX 2.9.1 [3, 4], based on interval contractors of the type developed in [1], to solve (5) on $[0, 2\pi]^5$. The input expressions contain only rational operations, square roots, sine, and cosine; all interval evaluations are outward rounded. Both the existence and uniqueness tests were active. The stopping tolerances for boxes and equalities were 10^{-10} .

Proposition 4.1. *The system $F(\theta) = 0$ has exactly 52 solutions in $[0, 2\pi]^5$. None lies on the boundary.*

Validated computational proof. The solver terminated successfully with 52 solution boxes and zero boundary, unknown, or pending boxes. Since the system is square, each returned solution box carries a uniqueness enclosure. Both the 52 existence boxes and the 52 larger uniqueness boxes were checked pairwise disjoint. Completeness of the cover, uniqueness within each box, and pairwise disjointness therefore imply that there are exactly 52 distinct roots. A second run with an independent random seed returned the same number of roots and the same graph certificate below. The two runs are summarized in Table 1. $\square$

TABLE 1. Independent complete interval runs. CPU time is diagnostic only.

seed	cells	solutions	boundary	unresolved	CPU seconds
1	175923	52	0	0	209.439
20260818	170765	52	0	0	206.732

The full boxes, rather than decimal representatives, are retained in the certificate. The largest coordinate width in either run is below $6 \cdot 10^{-14}$; this small width is not used as a substitute for interval containment.

5. THE POSSIBLE-ORTHOGONALITY GRAPH

Let $X_1, \dots, X_{52}$ be the certified solution boxes. For a pair $r \neq s$, interval arithmetic evaluates

$$(6) \quad Q_{rs} = \left| 1 + \sum_{j=1}^5 e^{i(\theta_{sj} - \theta_{rj})} \right|^2$$

over the entire box product $X_r \times X_s$. Define a graph G^+ on the 52 boxes by retaining rs as an edge exactly when the resulting interval contains zero.

Lemma 5.1. *The true orthogonality graph of the common-unbiased rays is a subgraph of G^+ .*

Proof. If the unique roots in X_r and X_s are orthogonal, then the exact value of (6) is zero. Outward-rounded interval evaluation over $X_r \times X_s$ must contain that value, so rs is retained. Thus deleting an edge is possible only after proving a strictly positive lower bound. $\square$

Proposition 5.2. *The graph G^+ has 48 edges, degree range 1–2, and no triangle.*

Finite interval certificate. All $\binom{52}{2} = 1326$ box pairs are evaluated. Zero is rigorously excluded for 1278 pairs. The smallest positive lower endpoint used in any exclusion is greater than $4.99 \cdot 10^{-5}$. The remaining 48 pairs are retained. A direct adjacency-bitset traversal finds no triangle. Repeating this calculation on the second complete interval cover gives the same counts. $\square$

Proof of Theorem 1.1. Suppose a third mutually unbiased basis existed. After fixing the phase of each column, its six columns would be six distinct solutions of (5). Orthonormality would make them a six-clique in the true orthogonality graph, hence in G^+ by Lemma 5.1. This is impossible because Proposition 5.2 says that G^+ is triangle-free. The same argument excludes even three pairwise orthogonal common-unbiased rays. $\square$

6. A BREAKER AT THE ENDPOINT $x = -1$

The validated calculation above was preceded by an exact countercheck at a simpler endpoint. It is useful both as a positive test of normalization and as a warning against promoting randomized parameter sampling to a decision theorem.

Let $\eta = e^{\pi i/6}$. Define K_- and L_- by exponentiating the entries of the following arrays modulo 12:

$$E_K = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 6 & 6 & 0 & 0 \\ 0 & 6 & 2 & 10 & 8 & 4 \\ 0 & 6 & 10 & 2 & 4 & 8 \\ 0 & 0 & 8 & 4 & 8 & 4 \\ 0 & 0 & 4 & 8 & 4 & 8 \end{pmatrix}, \quad E_L = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 3 & 3 & 9 & 9 & 9 \\ 3 & 7 & 11 & 1 & 5 & 9 \\ 7 & 3 & 11 & 1 & 9 & 5 \\ 0 & 4 & 8 & 4 & 8 & 0 \\ 4 & 0 & 8 & 4 & 0 & 8 \end{pmatrix},$$

so that $(K_-)_{jk} = \eta^{(E_K)_{jk}}$ and similarly for L_- . The first matrix is the $x = -1$ specialization of the same family.

Proposition 6.1. *The normalized matrices I_6 , $K_-/\sqrt{6}$, and $L_-/\sqrt{6}$ form a triple of mutually unbiased bases.*

Proof. Reduction in $\mathbb{Z}[X]/(X^4 - X^2 + 1)$ gives

$$K_-^* K_- = L_-^* L_- = 6I_6, \quad |(K_-^* L_-)_{jk}|^2 = 6 \quad (0 \leq j, k \leq 5).$$

These identities are also checked by a short exact-integer verifier. □

7. REPRODUCIBILITY AND SCOPE

The accompanying archive contains the exact radical generator for K , the five-equation IBEX input, the two complete solver covers, the solver logs, the C++ interval graph verifier, and the independent cyclotomic verifier for Proposition 6.1. A single script regenerates the equations, runs both complete solves, rebuilds the postprocessor, and verifies all claims. The postprocessor checks five variables, five equalities, successful solver status, the absence of all unresolved box classes, and pairwise disjointness of the 52 existence and uniqueness boxes. It then recomputes every interval in (6); stored decimal approximations are never trusted.

The trust boundary includes IBEX 2.9.1 and its interval backend. The COV file does not contain a separately replayable trace for every pruned branch, so the postprocessor alone is not a proof-carrying checker of the global solve. The entire solve is instead independently rerunnable from the displayed exact system.

The theorem is local to the matrix (1). It does not classify the full one-parameter family and does not address whether four MUBs exist in $\mathbb{C}^6$. It has not been translated into Lean or another proof assistant.

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